# Eric Gu

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## Education

**University of California, San Diego** June 2027
BS/MS in Computer Engineering **GPA:** 3.8 | **Major GPA:** 4.0

- **Relevant Coursework:** DSA, Operating Systems, Software Engineering, Deep Learning Models, Reinforcement Learning

## Experience

**Incoming Software Development Engineer Intern** September 2026
Amazon – Irvine, CA – Device OS

**Software Engineering Intern** June 2026 – September 2026
Viasat – Carlsbad, CA

- Built an LLM operations assistant using Google's ADK and Gemini (Vertex AI): a coordinator agent delegating to four MCP sub-agents and a Vertex AI Search base over 14,000+ Confluence pages.
- Integrated four MCP servers via ADK's McpToolset with least-privilege access (43-tool read-only GitHub allowlist) and pulled all runtime secrets from HashiCorp Vault via keyless GCP-IAM auth.
- Authored custom Grafana tools that interpolate BigQuery-backed panel SQL for live metrics, and deployed the agent as a private Cloud Run service with a FastAPI and single-page chat UI.

**Software Development Intern** June 2023 – September 2025
MaxNorm LLC – Jersey City, NJ

- Researched cryptocurrency markets and blockchain protocols, analyzing trends with Python technical and fundamental methods to produce reports that informed web3 investment decisions.
- Developed and deployed Solidity smart contracts for DeFi prototypes on Ethereum testnets, reducing security vulnerabilities by 40% through rigorous simulation testing.
- Backtested quantitative trading strategies with Pandas and NumPy, evaluating Sharpe ratio and drawdown on historical data to refine algorithms reaching up to 25% simulated returns.

## Projects

**VistaSR** | *Python, PyTorch, Diffusion Models* github.com/fordatab/sr

- Developed a diffusion-based super-resolution model (SR3) enabling 2x upscaling of landscape images, outperforming Real-ESRGAN in qualitative evaluations.
- Engineered a refined U-Net architecture with residual blocks and self-attention, enabling efficient training on mid-tier hardware to iteratively denoise inputs.
- Conducted quantitative assessments using PSNR/SSIM metrics and visual comparisons against FSRCNN, resulting in a robust solution for real-world applications.

**TriStage Compiler** | *Java, ANTLR* github.com/fordatab/tristage

- Built a compiler for a custom imperative language; implemented an ANTLR front-end for precise tokenization, error reporting, and LL(1) grammar resolution.
- Optimized the middle-end via static analysis and dead code elimination, reducing dynamic instruction counts by 50% over unoptimized IR.
- Engineered a MIPS32 back-end with intra-block greedy register allocation and liveness analysis, yielding 30-60% fewer memory loads.

**CUDA-NN** | *C++, CUDA* github.com/fordatab/gemm

- Beat PyTorch's conv stack by 26% (69 vs 94 ms) with a per-layer dispatcher sending the first layer to a custom CUDA kernel and the rest to cuDNN, at FP32 parity with autotuning on.
- Won the single-channel first layer against autotuned cuDNN (37 vs 45 ms) with a kernel emitting all 16 output channels from one shared-memory input load.
- Cut conv forward time 3x over an im2col + cuBLAS baseline across five kernel generations, in a VGG-style CNN and training framework built from scratch in C++/CUDA/Eigen.

## Skills

- **Languages:** JavaScript, TypeScript, Python, Java, SQL, Solidity
- **Front-End:** React, React Native, Redux, Tailwind CSS
- **Back-End:** Node.js, Express.js, Supabase, PostgreSQL, Vector Embeddings
- **Cloud & Tools:** AWS S3, Heroku, Docker, GitHub Actions, Git